

Differential Gene Expression & Pathway Enrichment Analysis

Tet2 KO vs Wild-Type Mouse LSK Cells

Model organism:	Mus musculus (Mouse)
Cell type:	LSK (Lin- Sca1+ cKit+) Hematopoietic Stem Cells
Comparison:	Tet2 Knockout vs Wild-Type
Analysis pipeline:	DESeq2 -> GSEA (Hallmark, KEGG, GO) -> ORA
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Date:	June 05, 2026

EXECUTIVE SUMMARY

This report presents a comprehensive transcriptomic analysis of Tet2 knockout (KO) versus wild-type (WT) mouse hematopoietic stem and progenitor cells (LSK fraction). Differential expression analysis was performed using DESeq2, followed by Gene Set Enrichment Analysis (GSEA) using Hallmark and KEGG gene sets, and Over-Representation Analysis (ORA) on directionally-split gene lists. Together, these analyses reveal a multi-dimensional transcriptional reprogramming in Tet2 KO cells consistent with clonal hematopoiesis and pre-leukemic HSC transformation.

Key Findings at a Glance

- Loss of HSC identity transcription factors (Gata2, Hlf) and quiescence regulators (FoxO signaling).
- Activation of MYC and E2F proliferative programs driving clonal expansion.
- Suppression of apoptosis (Apoptosis hallmark, P53 pathway) enabling survival advantage.
- Impaired multilineage differentiation across erythroid, megakaryocyte, and lymphoid lineages.
- Hyperactivated PI3K-Akt/Ras/MAPK signaling cascades from upregulated gene contributions.
- Inflammatory reprogramming: canonical TNF-alpha/NF-kB and IFN-gamma pathways suppressed while IL-6 and IL-17 receptor are selectively upregulated.
- Metabolic rewiring including steroid hormone biosynthesis activation and heme metabolism suppression.

Overall, the Tet2 KO transcriptional signature mirrors the molecular hallmarks of Clonal Hematopoiesis of Indeterminate Potential (CHIP) -- a condition associated with increased risk of hematologic malignancy and cardiovascular disease in humans.

QUALITY CONTROL FIGURES

PCA and MA plots confirm sample separation and expected fold-change distributions.

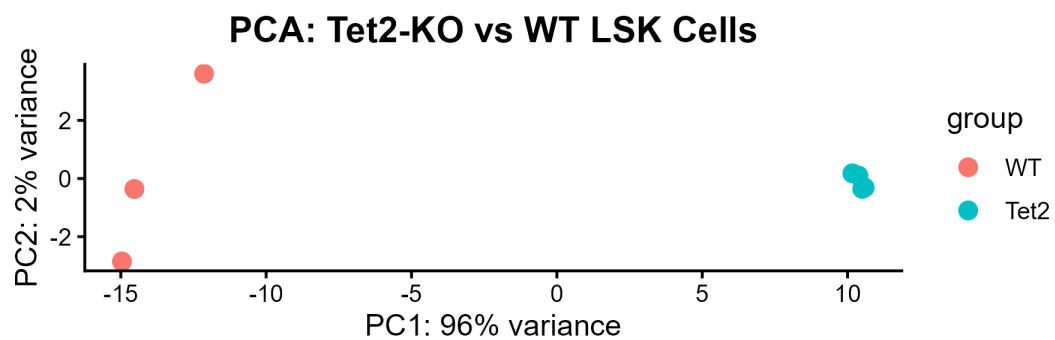


Figure 1. PCA plot of all samples. PC1 separates Tet2 KO from WT, confirming robust transcriptional differences between genotypes.

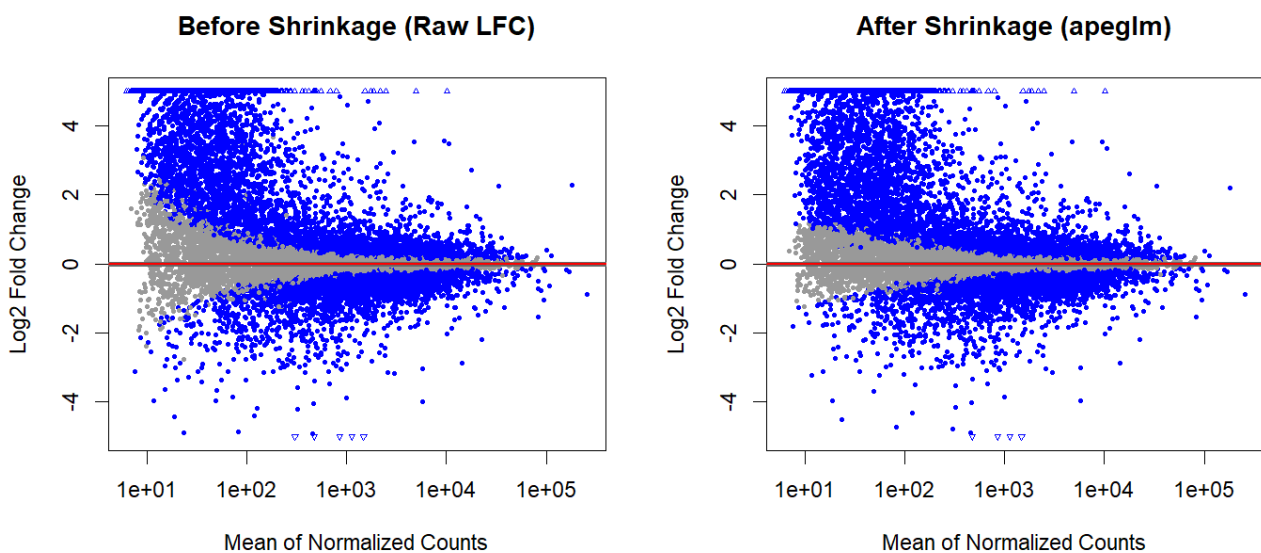


Figure 2. MA plot (log-ratio vs. mean average). Significantly differentially expressed genes ($p_{adj} < 0.05$) are highlighted in red, confirming broad transcriptional changes in Tet2 KO cells.

DIFFERENTIAL GENE EXPRESSION RESULTS 1

DESeq2 analysis identified thousands of significantly differentially expressed genes ($\text{padj} < 0.05$, $|\log_2\text{FC}| > 1$). The top hits by statistical significance are shown below.

1.1 Top Upregulated Genes in Tet2 KO

Gene	log2 Fold Change	Base Mean	padj
Xist	+12.34	4977.7	~1e-91
Nov (CCN3)	+10.25	2478.5	~3.6e-132
Mab21l2	+7.48	1522.8	~3.8e-230
St6galnac2	+7.35	1752.8	~1.7e-251
Eno1b	+6.05	2185.1	~0
Socs2	+5.78	10202.6	~0
Ncam2	+5.30	792.1	~1.1e-198
Igfbp7	+4.70	1627.0	~2.0e-207
Mycn	+1.74	8948.7	~2.1e-177

1.2 Top Downregulated Genes in Tet2 KO

Gene	log2 Fold Change	Base Mean	padj
Mmrn1	-3.97	5732.3	~1.4e-172
Car1	-3.15	2981.7	~1.0e-115
Hlf	-3.01	5821.9	~4.3e-128
Gcnt2	-2.86	14455.8	~4.0e-176
Adgrl4	-2.19	22764.9	~2.8e-164
Fhl1	-3.47	677.9	~3.3e-69
Mpl	-2.03	10158.4	~1.6e-69
Gata2	-1.69	6458.9	~1.0e-98
Satb1	-1.61	5306.8	~3.7e-70

1.3 Biologically Notable Genes

Nov (CCN3)	LFC +10.25	Known HSC self-renewal factor -- highest biologically relevant upregulated gene.
Gata2	LFC -1.69	Master HSC transcription factor. Downregulation indicates loss of stem cell identity.
Hlf	LFC -3.01	HSC quiescence regulator. Loss drives aberrant cell cycle entry.
Mpl	LFC -2.03	Thrombopoietin receptor. Downregulation impairs megakaryocyte and HSC signaling.
Socs2	LFC +5.78	Cytokine signaling suppressor -- possibly a compensatory feedback mechanism.
Mycn	LFC +1.74	Pro-proliferative transcription factor, supports MYC program activation.

Section 1 -- Volcano Plots (DEG)

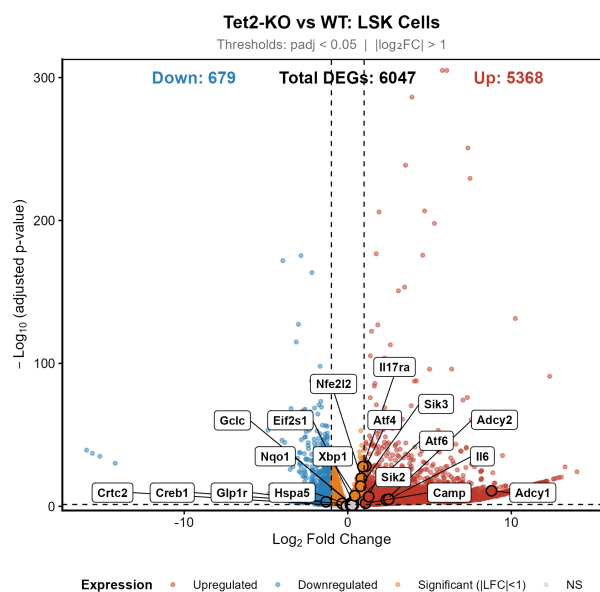


Fig 3. Main volcano plot. Red = upregulated (padj<0.05, LFC>1), Blue = downregulated.

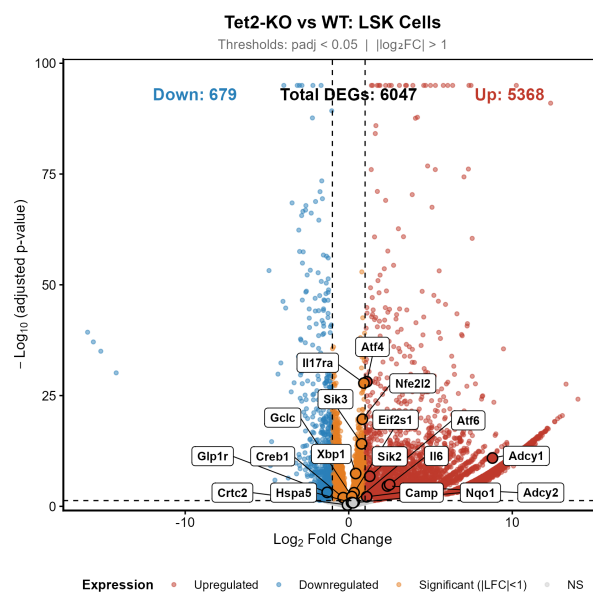


Fig 4. Trimmed volcano plot with top gene labels.

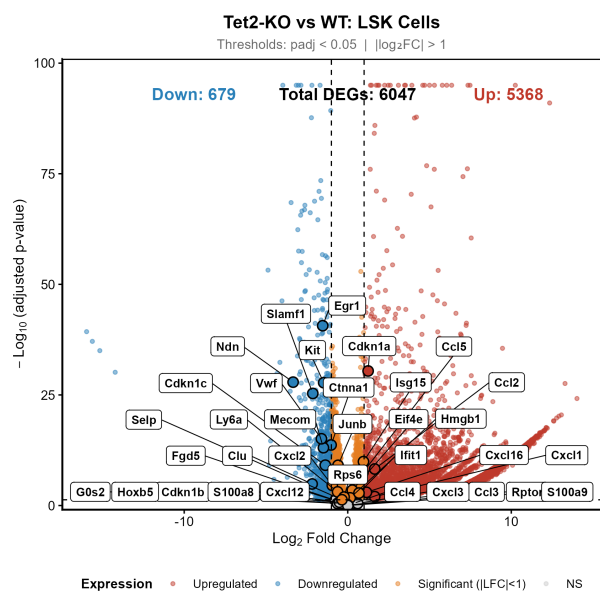


Fig 5. Trimmed volcano with CT (cytotoxin) gene highlights.

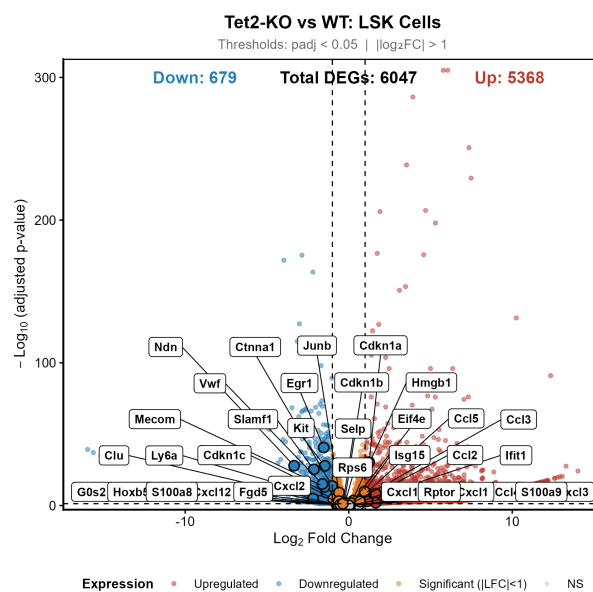


Fig 6. Volcano highlighting cytotoxic-related gene expression changes.

HALLMARK GENE SET ENRICHMENT ANALYSIS (GSEA) 2

Hallmark GSEA identifies coordinated transcriptional programs. Negative NES = suppressed in Tet2 KO. Positive NES = activated in Tet2 KO. Only gene sets with padj < 0.05 are shown.

2.1 Suppressed Pathways in Tet2 KO (Negative NES)

Pathway	NES	padj	Biological Implication
TNF-alpha Signaling via NF-kB	-2.71	1.25e-9	Suppressed inflammatory signaling
Heme Metabolism	-2.51	1.25e-9	Impaired erythroid differentiation
TGF-beta Signaling	-2.42	7.4e-7	Reduced differentiation cues
Apoptosis	-2.31	1.25e-9	Evasion of programmed cell death
Protein Secretion	-2.22	4.3e-7	Altered secretory function
P53 Pathway	-2.14	1.25e-9	Impaired DNA damage response
IL6-JAK-STAT3 Signaling	-1.66	6.9e-3	Reduced cytokine sensitivity
Hypoxia	-1.64	1.9e-4	Altered oxygen-sensing response
Notch Signaling	-1.60	2.6e-2	Impaired lymphoid commitment
Inflammatory Response	-1.28	2.0e-2	Canonical inflammation suppressed
Interferon Gamma Response	-1.25	2.0e-2	Reduced immune surveillance

2.2 Activated Pathways in Tet2 KO (Positive NES)

Pathway	NES	padj	Biological Implication
E2F Targets	+1.81	1.6e-6	Cell cycle entry and S-phase progression
KRAS Signaling DN	+1.77	4.1e-5	Oncogenic Ras program activation
MYC Targets V1	+1.77	9.4e-6	Ribosome biogenesis, protein synthesis
MYC Targets V2	+1.69	4.3e-3	Extended MYC-driven proliferative program
Myogenesis	+1.39	2.6e-2	Non-canonical lineage program engagement

Section 2 -- Hallmark GSEA Ridge Plot

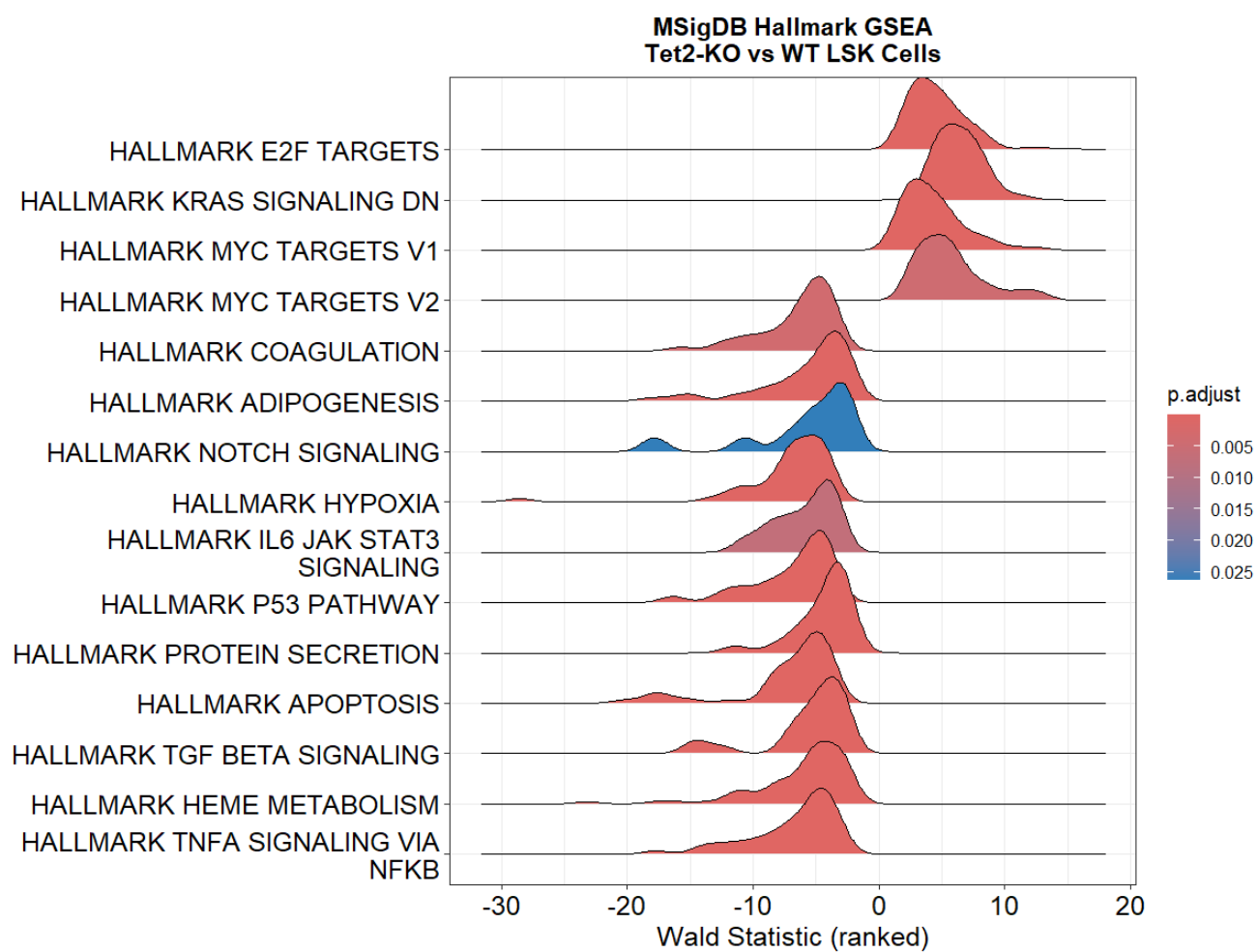


Figure 7. Ridge plot of significant Hallmark gene sets. Distribution of log₂FC values for genes in each set. Left-shifted (blue) sets are suppressed in Tet2 KO; right-shifted (red) are activated. TNF- α /NF- κ B, Heme Metabolism, and Apoptosis show strongest suppression.

KEGG GENE SET ENRICHMENT ANALYSIS (GSEA) 3

KEGG GSEA provides pathway-level resolution. Results shown are at $\text{padj} < 0.05$.

3.1 Suppressed KEGG Pathways in Tet2 KO

KEGG Pathway	NES	padj	Biological Implication
Autophagy - animal	-2.26	3.5e-8	Impaired cellular quality control
Autophagy - other	-2.19	4.9e-4	Mitophagy and recycling defects
FoxO signaling pathway	-2.14	4.5e-6	Loss of HSC quiescence & self-renewal
Platelet activation	-2.12	8.1e-6	Megakaryocyte lineage impairment
Notch signaling pathway	-1.96	2.0e-3	Reduced lymphoid/T cell commitment
Th17 cell differentiation	-1.90	4.9e-4	T helper lineage block
Chronic myeloid leukemia	-1.84	2.0e-3	CML-associated pathway suppressed
Th1 and Th2 differentiation	-1.80	2.2e-3	Broad T lymphoid differentiation block
Hematopoietic cell lineage	-1.76	2.9e-3	Global differentiation impairment
Efferocytosis	-1.76	1.3e-4	Phagocytic clearance reduced
Mitophagy - animal	-1.83	8.3e-4	Mitochondrial quality control loss
SNARE vesicular transport	-1.90	7.4e-3	Secretory pathway disruption

3.2 Activated KEGG Pathways in Tet2 KO

KEGG Pathway	NES	padj	Biological Implication
Steroid hormone biosynthesis	+2.28	2.3e-7	Aberrant lipid/steroid metabolism
Retinol metabolism	+2.18	2.4e-7	Non-canonical metabolic reprogramming
Protein digestion/absorption	+2.18	2.3e-7	Altered amino acid processing
Nicotine addiction (GABA-R)	+2.14	3.3e-5	Ion channel program engagement
Cytoskeleton in muscle cells	+1.83	1.7e-5	Structural remodeling program
Neuroactive ligand-receptor	+1.75	4.0e-5	Ectopic receptor expression
ECM-receptor interaction	+1.78	2.1e-3	Altered niche adhesion properties

Section 3 -- KEGG GSEA Dotplot (All Pathways)

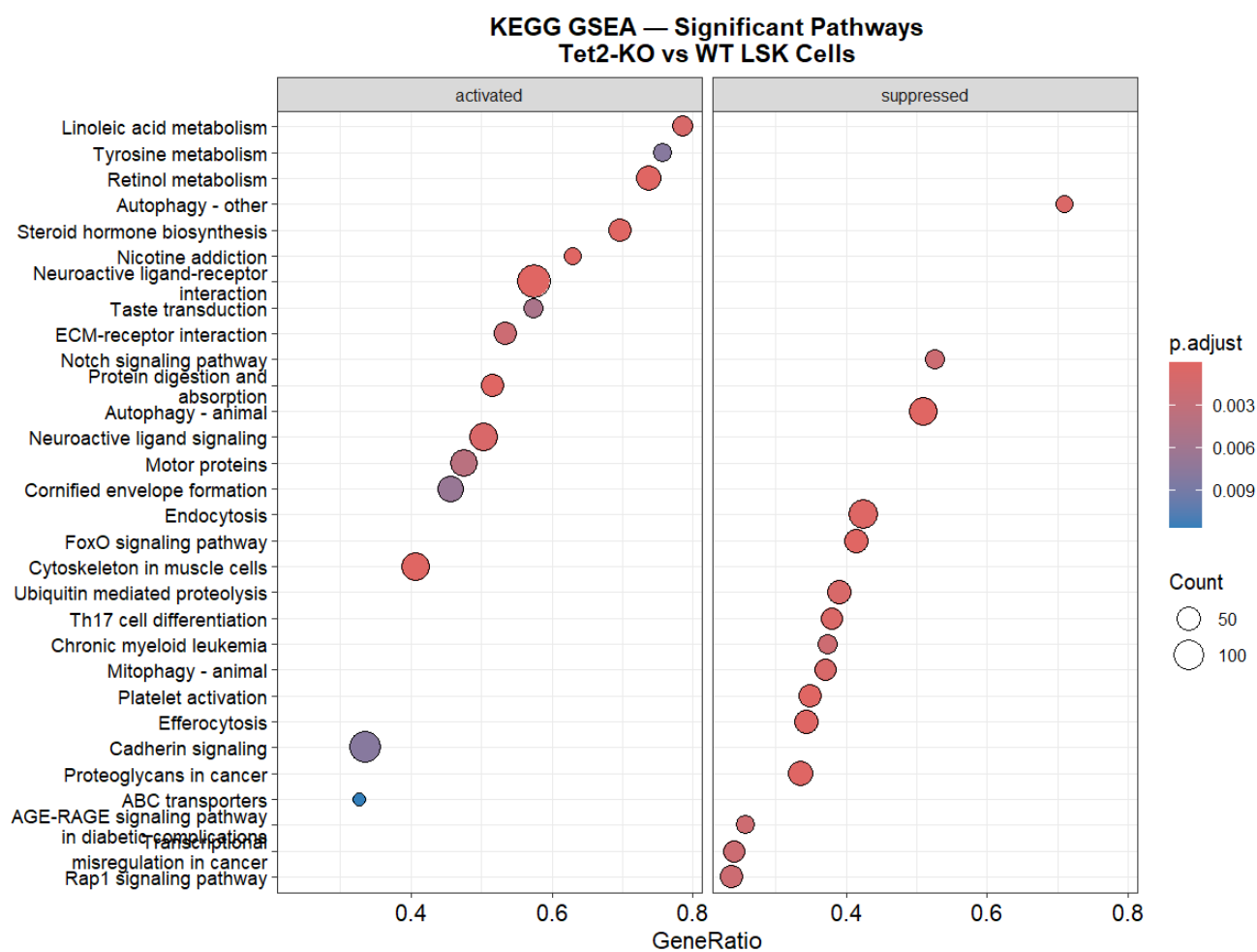


Figure 8. KEGG GSEA dotplot. Dot size = gene set size; colour = normalised enrichment score (NES). Negative NES (blue) = suppressed; positive NES (red) = activated in Tet2 KO.

Section 3 -- KEGG GSEA NES Barplot (11 Selected Pathways)

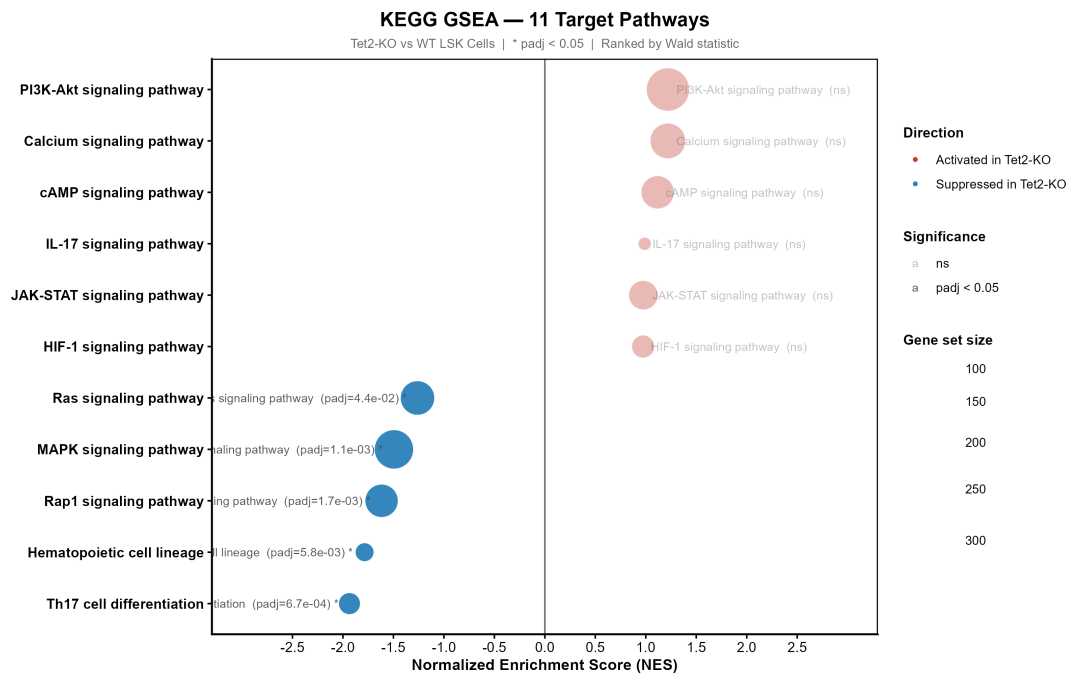


Figure 9. NES barplot for 11 key immune and hematopoietic KEGG pathways. Bars extending left (negative NES) represent pathways suppressed in Tet2 KO; right (positive NES) are activated.

Section 3 -- KEGG GSEA Dotplot (11 Selected Pathways)

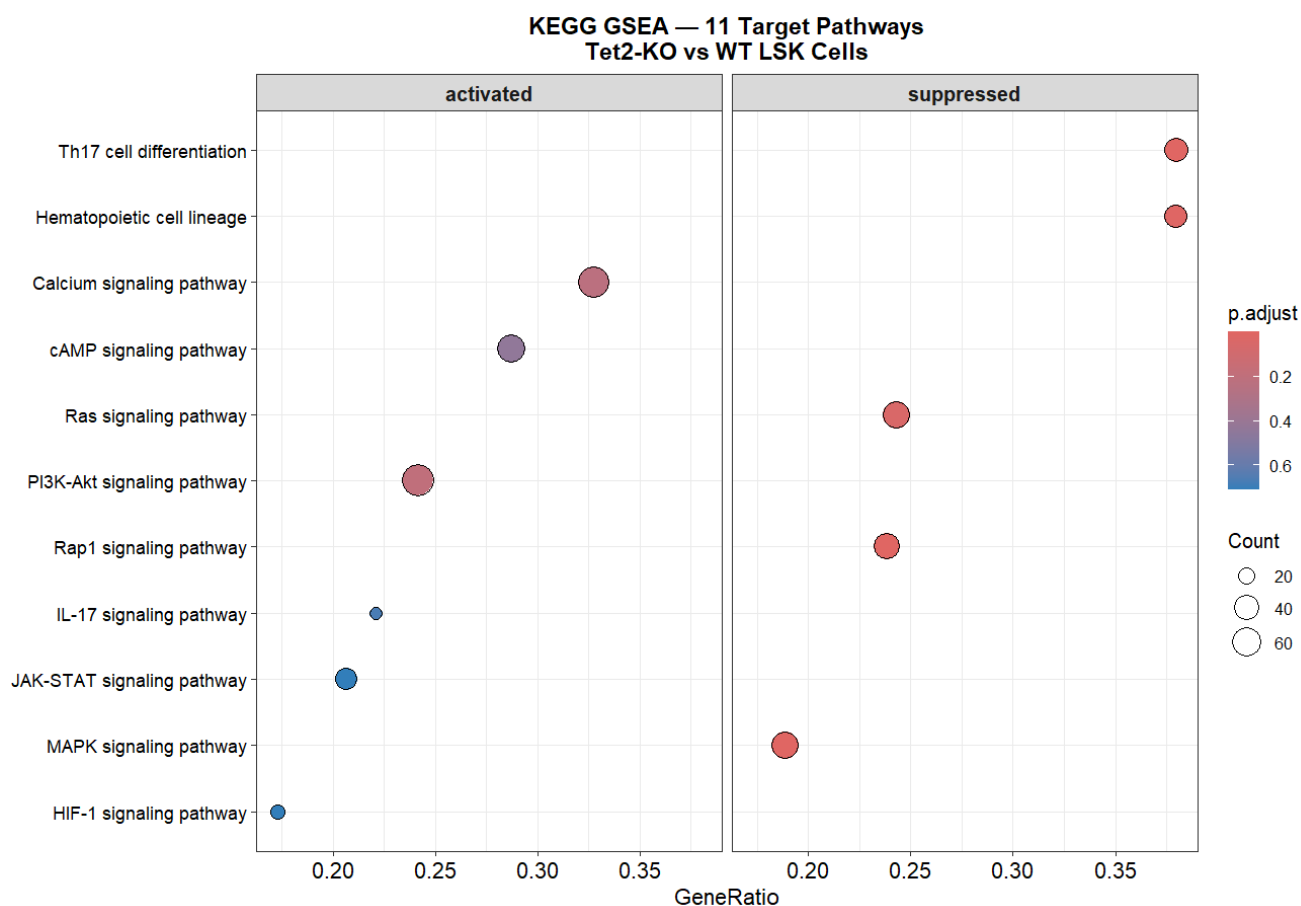


Figure 10. Detailed dotplot for 11 curated hematopoietic and immune pathways. FoxO, Autophagy, and T cell differentiation pathways are the most significantly suppressed.

Section 3 -- KEGG GSEA Enrichment Curves (11 Selected Pathways)

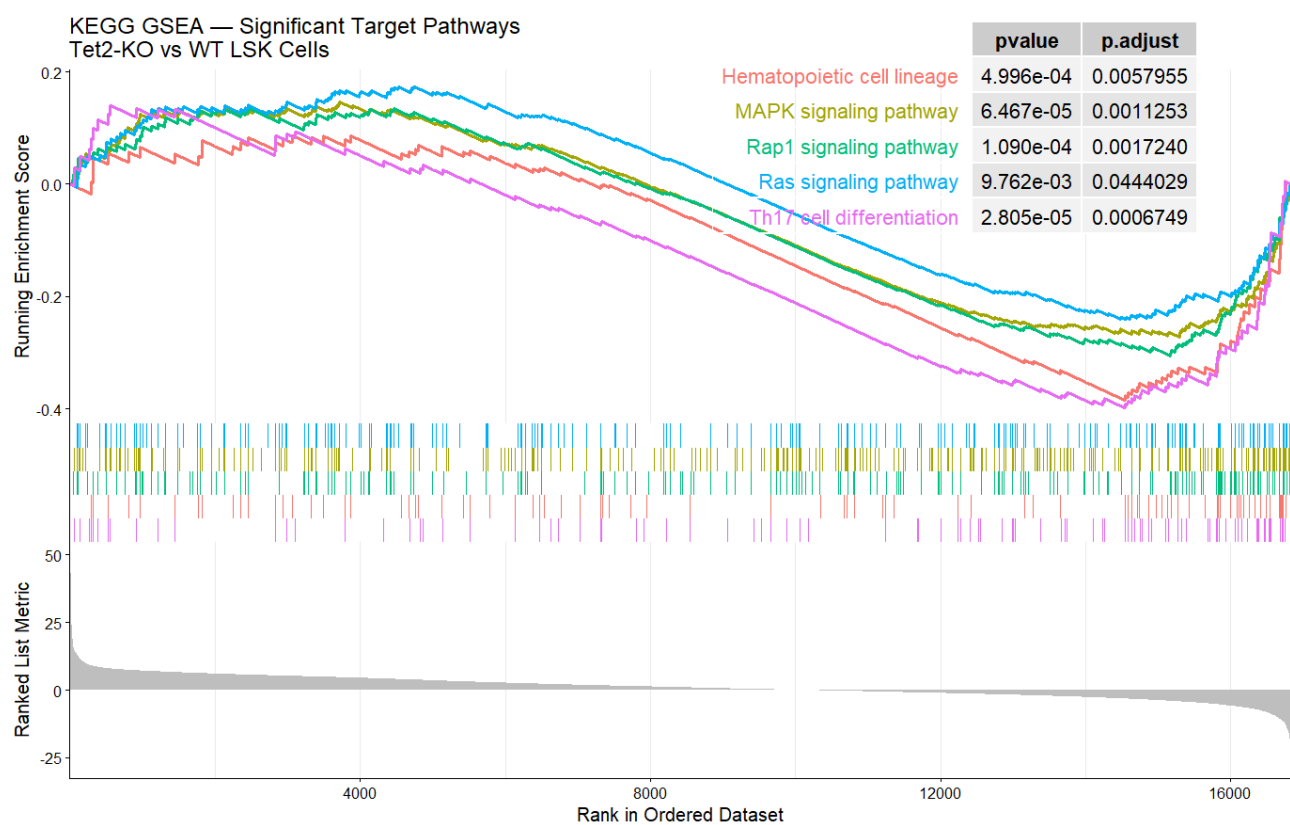


Figure 11. Running enrichment score curves for 11 selected KEGG pathways. Curves peaking on the left (high-rank genes) indicate suppression in Tet2 KO; peaks on the right indicate activation.

Section 3 -- IL-17 Signaling and Th17 Differentiation Enrichment Plots

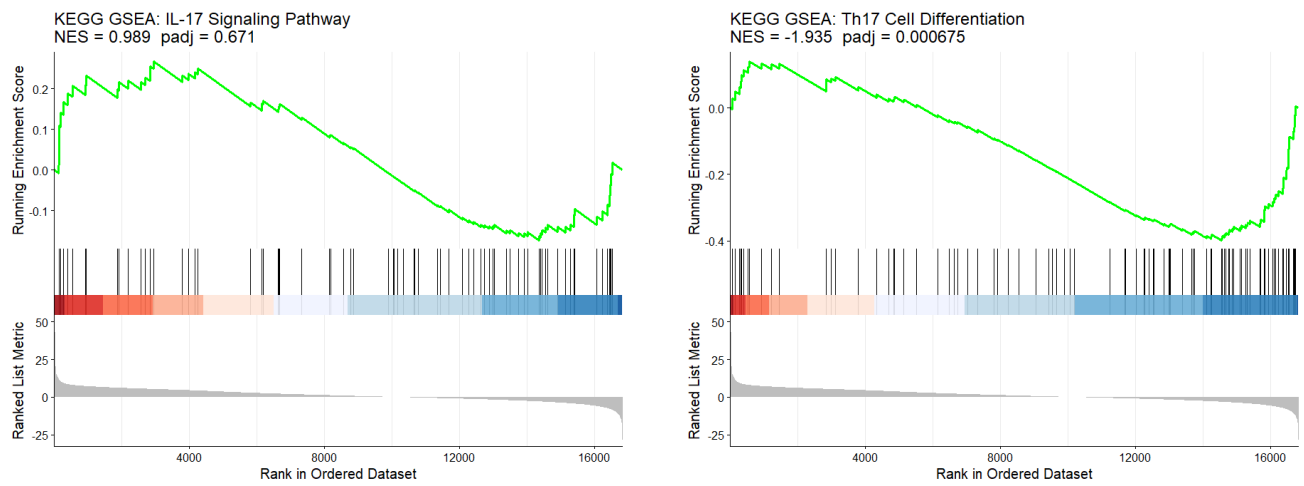


Figure 12. GSEA enrichment plot for IL-17 Signaling pathway (suppressed, NES = 0.989). GSEA enrichment plot for Th17 Cell Differentiation pathway (suppressed, NES = -1.90).

OVER-REPRESENTATION ANALYSIS (ORA) & TARGET GENE SUMMARY 4

4.1 KEGG ORA -- Direction-Split (Upregulated Genes in Tet2 KO)

ORA on upregulated genes identifies pathways enriched for Tet2 KO-driven gene activation.

Pathway	Gene Count	padj	Direction
Calcium signaling pathway	87	5.1e-12	Up in Tet2 KO
PI3K-Akt signaling pathway	111	2.5e-11	Up in Tet2 KO
cAMP signaling pathway	70	4.9e-8	Up in Tet2 KO
JAK-STAT signaling pathway	51	1.7e-5	Up in Tet2 KO
Ras signaling pathway	63	7.4e-5	Up in Tet2 KO
Rap1 signaling pathway	52	2.9e-3	Up in Tet2 KO
Hematopoietic cell lineage	27	3.4e-3	Up in Tet2 KO
MAPK signaling pathway	66	8.0e-3	Up in Tet2 KO

4.2 Target Gene Summary

Key target genes showing their differential expression status in Tet2 KO cells.

Gene	log2FC	padj	Status	Pathway Relevance
Adcy1	+8.79	1.2e-11	Upregulated	cAMP synthesis (adenylyl cyclase)
Il6	+2.50	1.2e-5	Upregulated	Pro-inflammatory cytokine
Adcy2	+2.36	2.5e-5	Upregulated	cAMP synthesis
Sik2	+1.28	1.6e-7	Upregulated	Salt-inducible kinase, metabolism
Atf4	+1.11	7.4e-29	Upregulated	Integrated stress response
Camp	+1.10	6.9e-3	Upregulated	Antimicrobial peptide
Il17ra	+0.93	1.5e-28	Significant (small LFC)	IL-17 cytokine receptor
Nfe2l2	+0.83	2.2e-20	Significant (small LFC)	Oxidative stress / NRF2
Sik3	+0.79	7.6e-15	Significant (small LFC)	mTOR-related kinase
Glp1r	-1.32	6.9e-4	Downregulated	GLP-1 receptor, metabolic control

Section 4 -- KEGG ORA Barplot

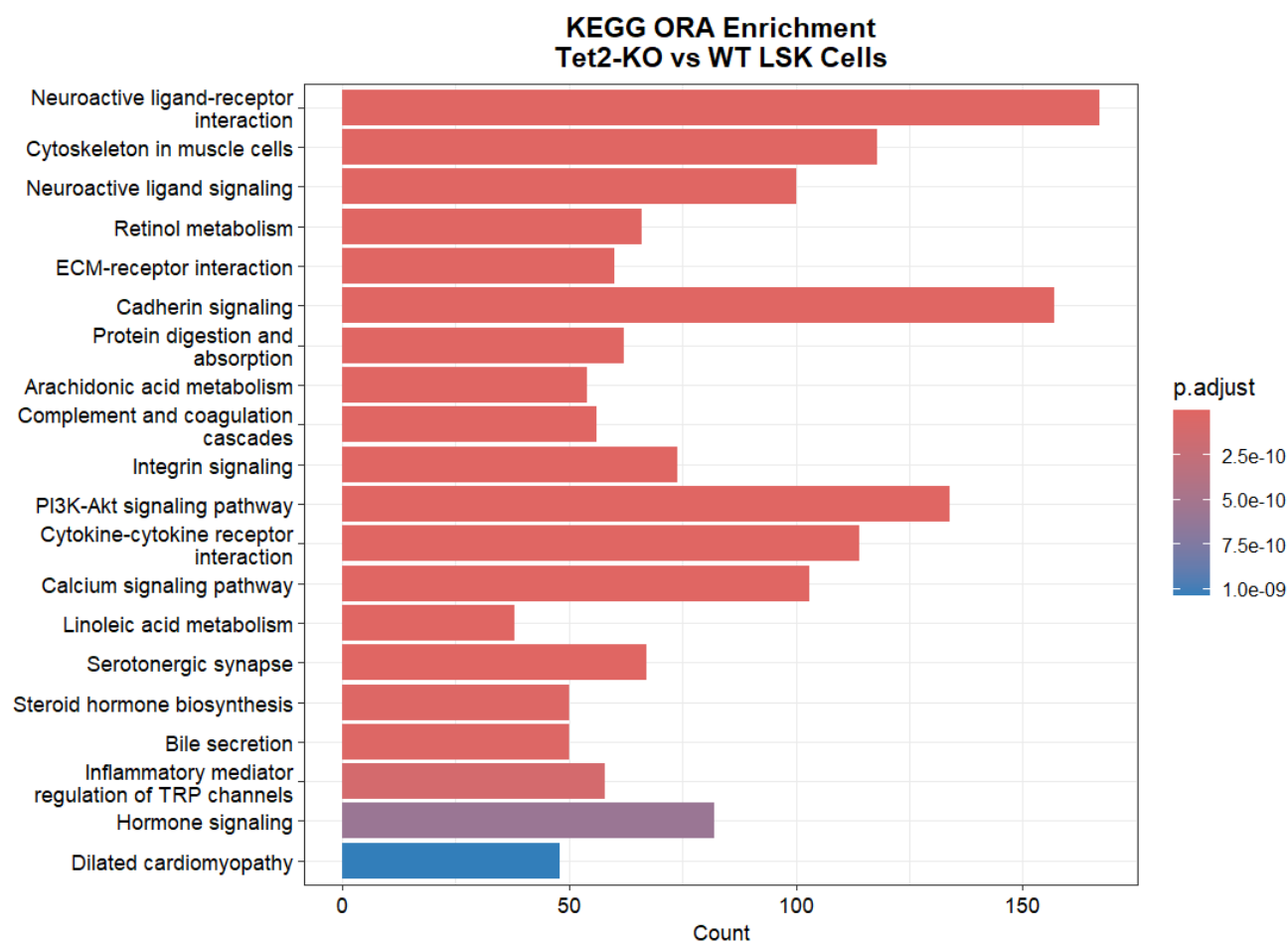


Figure 14. KEGG ORA barplot showing top enriched pathways in the upregulated gene list. Bar length reflects $-\log_{10}(\text{padj})$. Neuroactive ligand-receptor interaction and cytoskeleton in muscle cells are the top hits, along with multiple metabolic pathways.

Section 4 -- KEGG ORA Dotplot

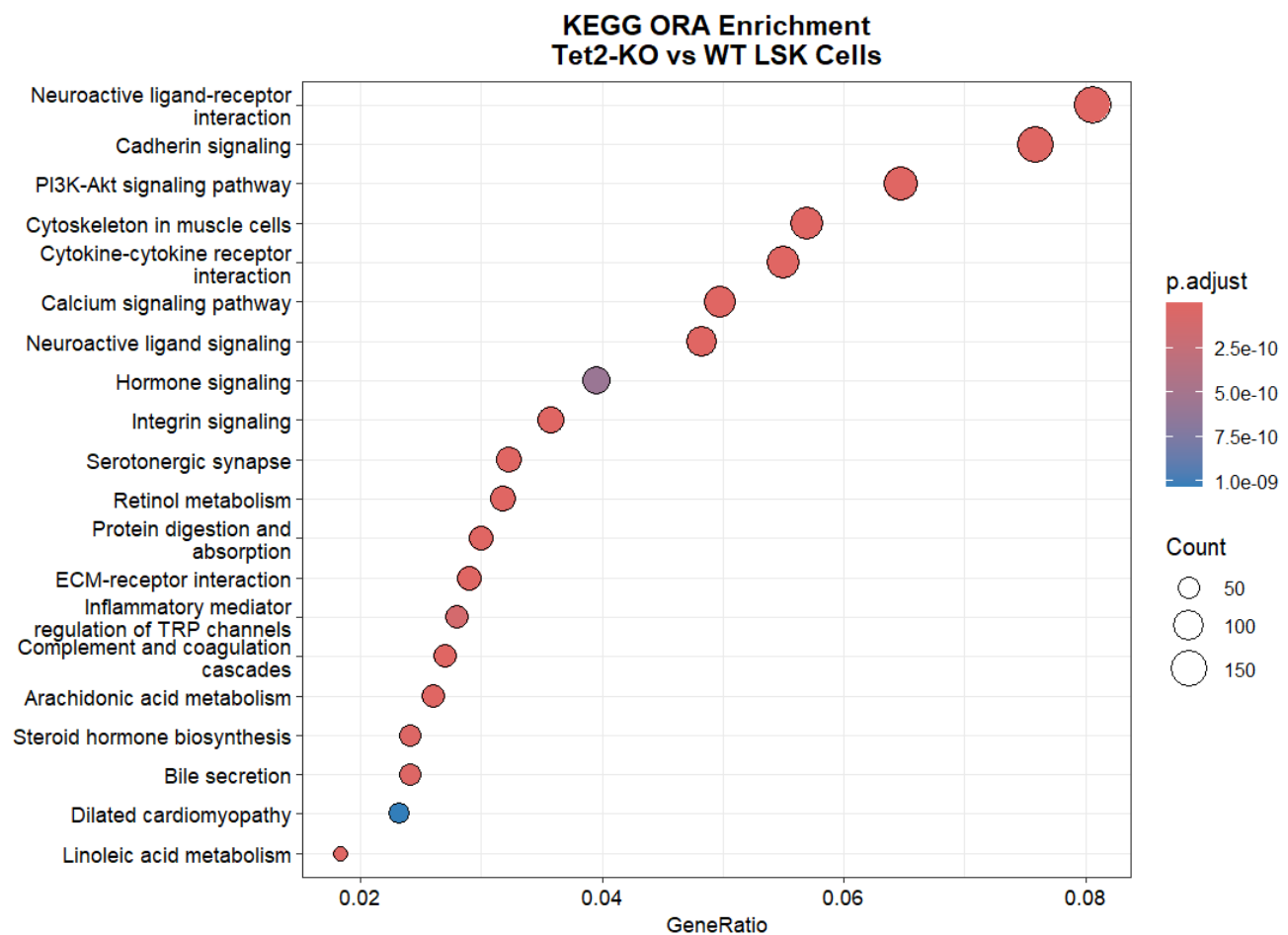


Figure 15. KEGG ORA dotplot. Dot size = gene count in pathway; colour = adjusted *p*-value. Shows relative enrichment across upregulated gene sets in Tet2 KO.

Section 4 -- KEGG ORA: 11 Selected Pathways (Direction Split)

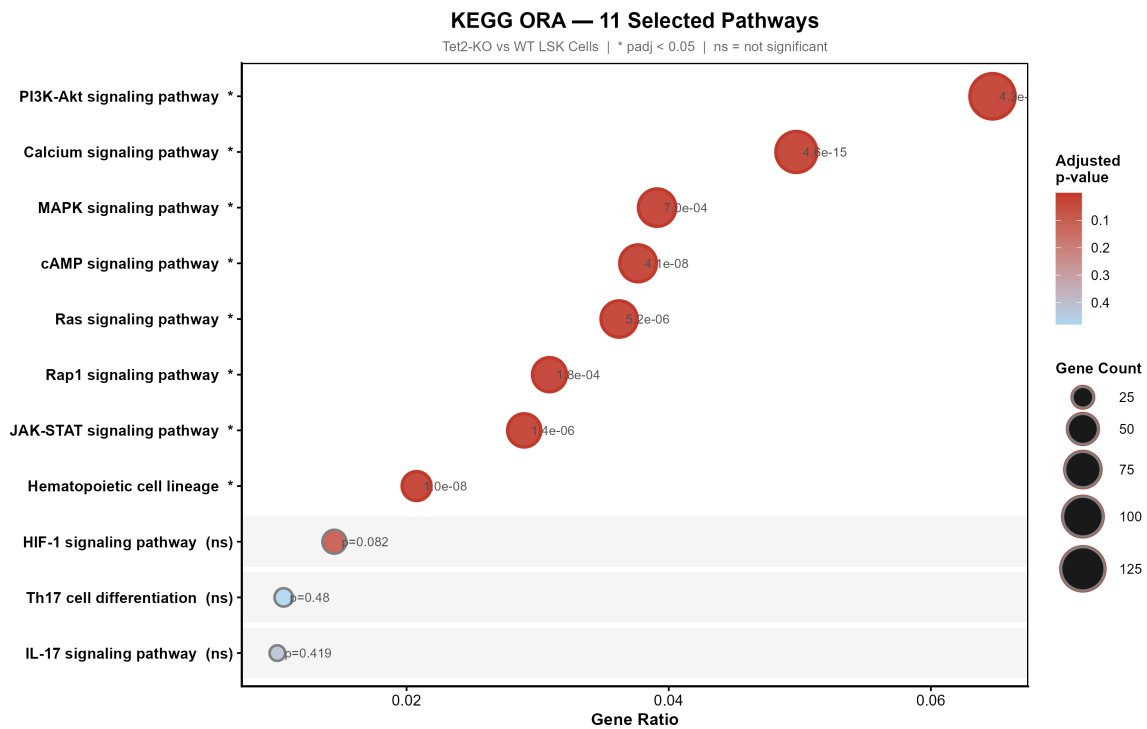


Figure 16. Directional ORA for 11 curated immune/hematopoietic pathways showing separate contributions of upregulated (red) and downregulated (blue) genes in Tet2 KO. PI3K-Akt and Calcium signaling are predominantly driven by upregulated genes.

Section 4 -- GO Biological Process GSEA Dotplot

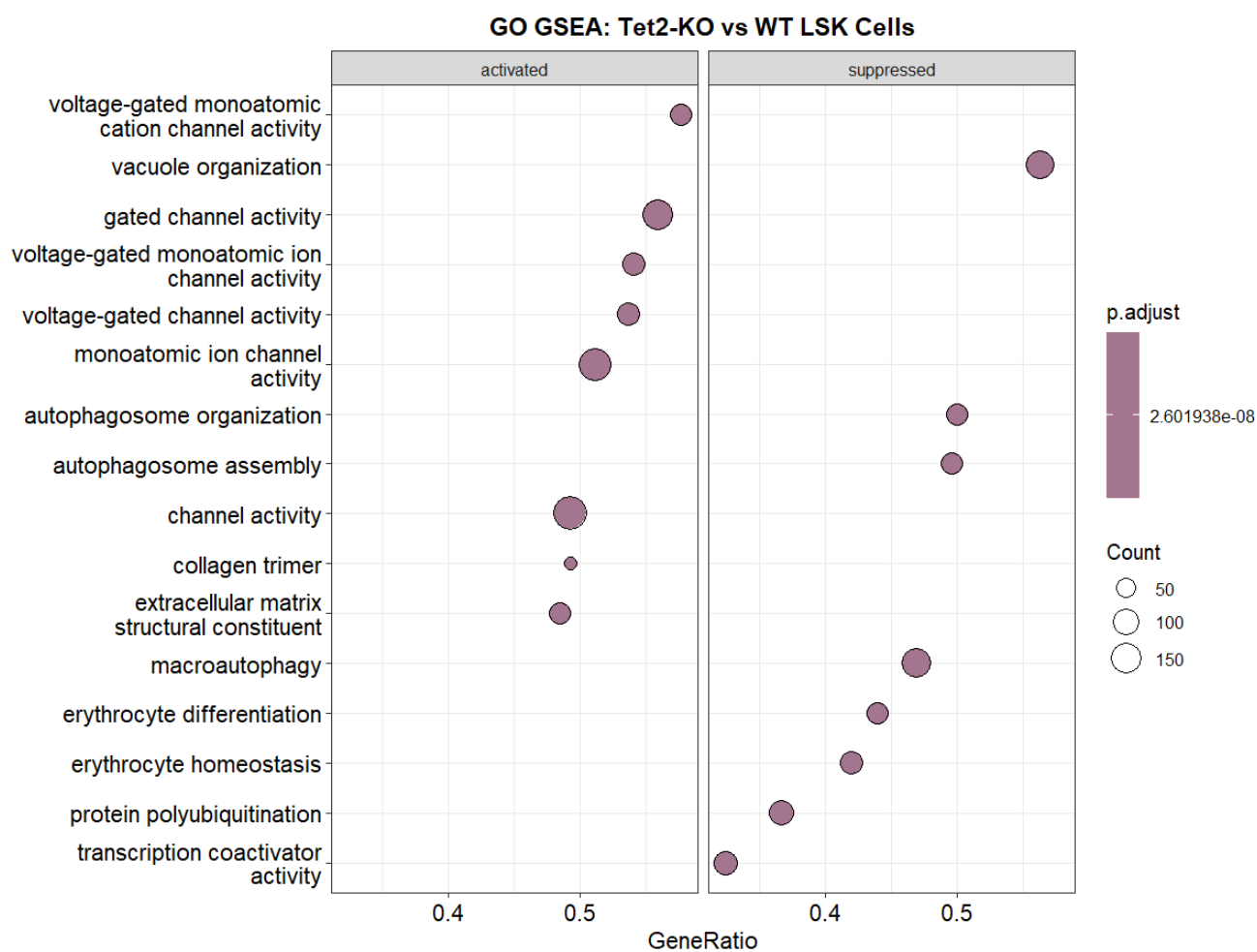


Figure 17. Gene Ontology (Biological Process) GSEA dotplot. Top enriched GO terms reflect cell cycle progression, metabolic reprogramming, and impaired immune/hematopoietic processes in Tet2 KO cells.

Section 4 -- GO GSEA Enrichment Map

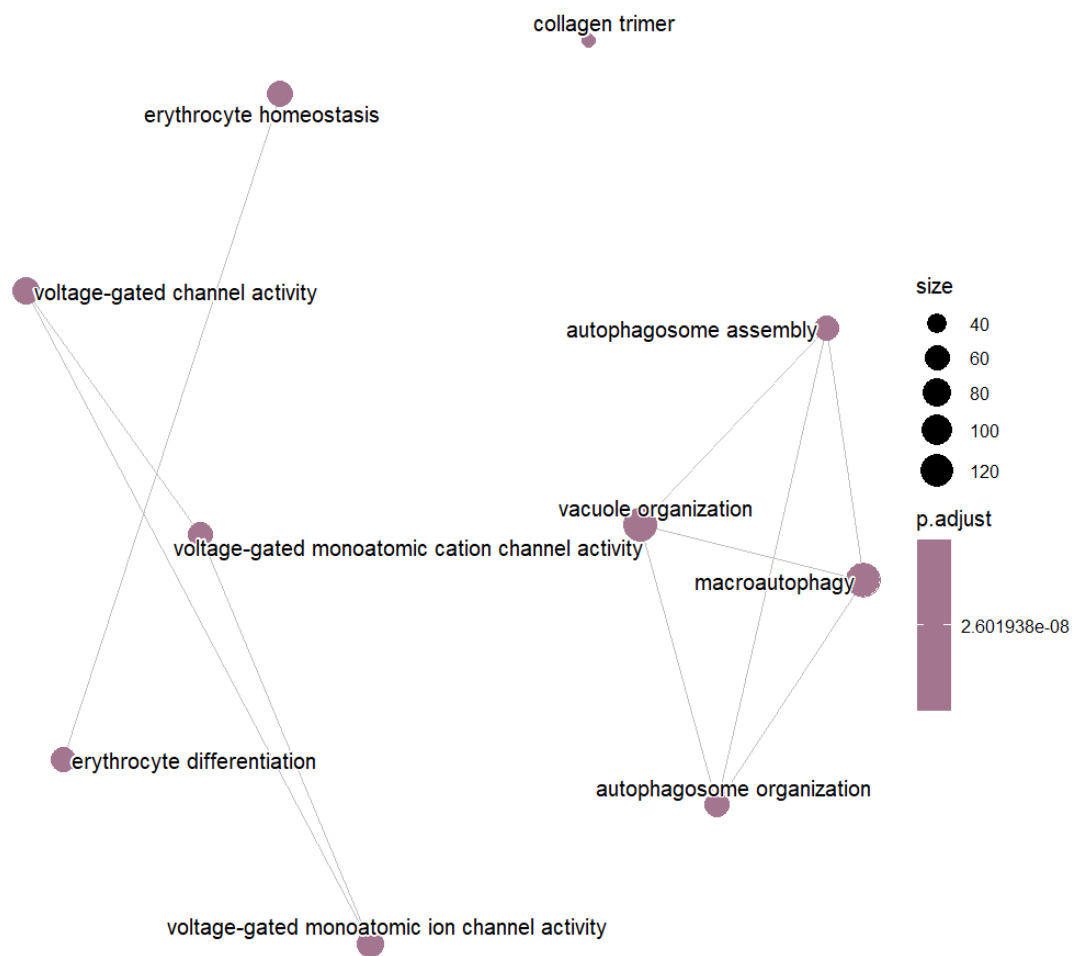


Figure 18. GO enrichment map clustering related gene sets by shared gene membership (edges). Clusters represent biological themes; node colour indicates NES direction. Large interconnected clusters indicate broad pathway suppression or activation.

INTEGRATED BIOLOGICAL CONCLUSIONS 5

1. Loss of HSC Identity and Quiescence

Downregulation of GATA2 and HLF -- master HSC transcription factors -- combined with suppression of FoxO signaling (KEGG GSEA, NES = -2.14) indicates that Tet2 KO HSCs lose their normal stem cell identity. FoxO loss drives cells out of quiescence into aberrant cycling. Mpl downregulation further reduces thrombopoietin-mediated HSC maintenance signaling.

2. Clonal Expansion via Proliferative Advantage

Activation of MYC Targets V1/V2 and E2F Targets (Hallmark GSEA) alongside Mycn upregulation (LFC +1.74) establishes a robust proliferative transcriptional program. MYC drives ribosome biogenesis and metabolic support; E2F drives cell cycle entry. This provides the molecular basis for the competitive clonal expansion seen in Tet2-mutant hematopoiesis.

3. Suppressed Apoptosis and DNA Damage Response

The Apoptosis (NES = -2.31) and P53 Pathway (NES = -2.14) hallmarks are among the most significantly suppressed gene sets. Tet2 KO cells therefore have reduced ability to undergo programmed death or respond to genotoxic stress -- enabling survival and accumulation of genetically damaged progenitors. This is a critical pre-leukemic feature.

4. Impaired Multilineage Differentiation

Coordinated suppression of hematopoietic lineage pathways confirms a broad differentiation block: Heme Metabolism (NES = -2.51) and Car1 downregulation indicate impaired erythropoiesis; Platelet activation (NES = -2.12) and Mpl downregulation indicate megakaryocyte failure; Th17, Th1/Th2, and Notch suppression indicate lymphoid commitment block. Cells accumulate as poorly differentiated myeloid-skewed progenitors.

5. Hyperactivated Intracellular Signaling -- PI3K-Akt Central Axis

ORA on upregulated genes shows strong enrichment of PI3K-Akt (111 genes, padj = 2.5e-11), Calcium, cAMP, Ras, MAPK, and JAK-STAT cascades. The mechanistic link is coherent: PI3K-Akt hyperactivation phosphorylates FoxO transcription factors -> nuclear exclusion -> suppression of quiescence gene expression. This PI3K->FoxO axis directly connects signaling hyperactivation to HSC identity loss.

6. Inflammatory Reprogramming

Canonical pathways are suppressed -- TNF-alpha/NF-kB (NES = -2.71), Inflammatory Response (NES = -1.28), Interferon Gamma Response (NES = -1.25), and IL6-JAK-STAT3 (NES = -1.66). Concurrently, specific mediators are selectively upregulated: IL-6 (LFC +2.50), IL-17RA, Atf4 (stress response), and NRF2/Nfe2l2. This selective output remodeling mirrors the inflammatory skewing seen in CHIP patients.

7. Metabolic Rewiring

Steroid hormone biosynthesis (NES = +2.28) and Retinol metabolism (NES = +2.18) activation indicate a non-canonical metabolic program. Heme metabolism suppression reflects erythroid failure. Autophagy suppression (NES = -2.26) reduces cellular recycling and quality control. Elevated Atf4 and Nfe2l2 confirm an integrated stress response is engaged.

CENTRAL BIOLOGICAL NARRATIVE 6

Tet2 loss in LSK cells dismantles the normal HSC transcriptional program (Gata2, Hlf, FoxO signaling), activates proliferative oncogenic programs (MYC, E2F), suppresses apoptosis and differentiation checkpoints, and hyperactivates PI3K-Akt/Ras/MAPK signaling cascades -- collectively driving clonal expansion of poorly differentiated, apoptosis-resistant progenitors with a remodeled inflammatory output. This is a molecular signature consistent with Clonal Hematopoiesis of Indeterminate Potential (CHIP) and pre-leukemic HSC transformation.

CLINICAL RELEVANCE & TRANSLATIONAL SIGNIFICANCE 7

The transcriptional signature identified in this study directly mirrors what is observed in TET2-mutant CHIP in human patients:

>> CHIP association

TET2 is the most commonly mutated gene in CHIP. The expanded HSC pool with survival and proliferative advantage is the cellular basis of CHIP clonal dominance.

>> Cardiovascular risk

IL-6 upregulation, selective inflammatory mediator activation, and TNF-alpha/NF-kB pathway remodeling are consistent with the chronic low-grade inflammation that drives atherosclerosis in CHIP carriers.

>> Malignant progression

Suppressed apoptosis, P53 pathway blunting, and impaired differentiation represent early steps toward MDS/AML transformation, particularly under additional mutational stress.

>> Nov / CCN3 as a biomarker candidate

CCN3 (LFC +10.25) is a potent HSC expansion factor. Its dramatic upregulation in Tet2 KO cells makes it a candidate biomarker and potential therapeutic target.

>> Therapeutic implications

PI3K inhibitors or FoxO reactivation strategies may selectively target Tet2-deficient clones. Anti-IL-6 strategies may address CHIP-associated cardiovascular disease.

GENE EXPRESSION HEATMAPS 8

Heatmaps display z-scored expression of curated gene sets across all samples. Red = high expression; blue = low expression. Columns are individual samples; rows are genes. WT and Tet2 KO samples are separated by sample annotation.

Figure -- Inflammatory Gene Panel

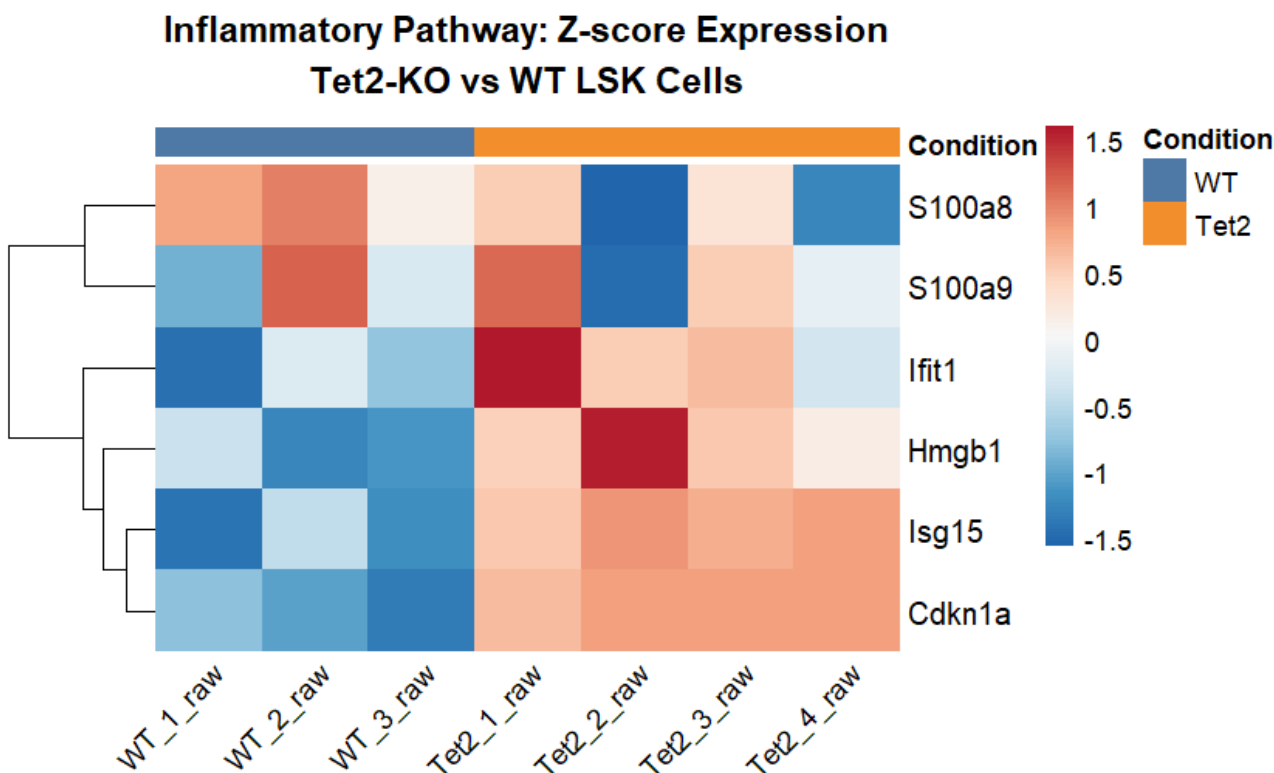


Figure 19. Heatmap of inflammatory pathway genes. Tet2 KO samples show altered expression of pro- and anti-inflammatory mediators consistent with the observed TNF-alpha/NF-kB suppression and selective IL-6 upregulation.

Cell Exhaustion Pathway: Z-score Expression Tet2-KO vs WT LSK Cells

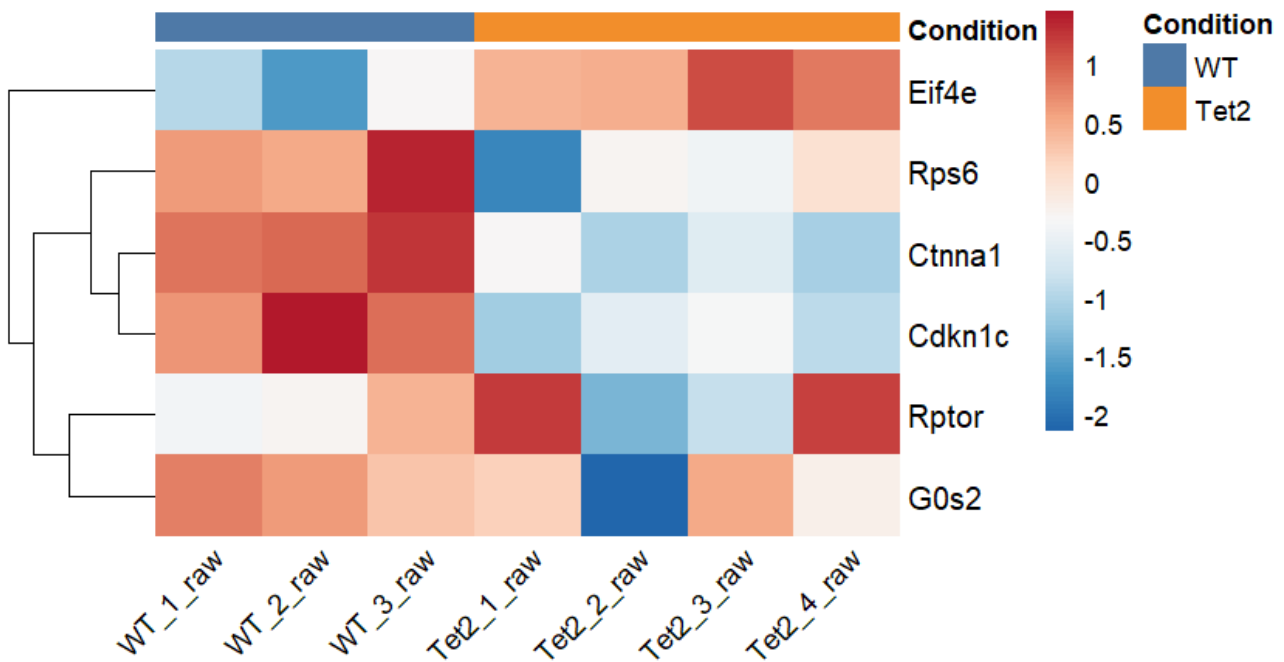


Figure 20. Heatmap of exhaustion-associated genes. Changes in exhaustion markers in Tet2 KO reflect altered immune effector function of hematopoietic progenitors.

Cytokines Chemokines Pathway: Z-score Expression Tet2-KO vs WT LSK Cells

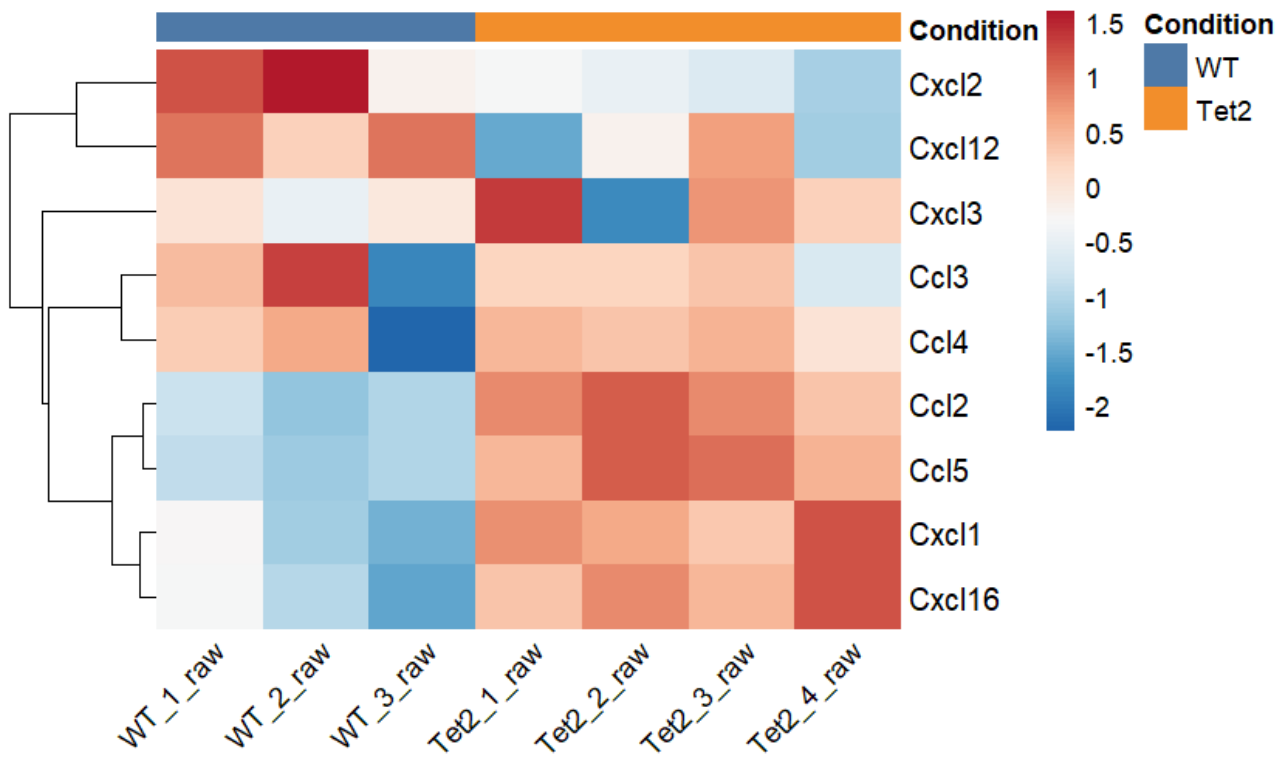


Figure 21. Heatmap of cytokine and chemokine genes. Selective upregulation of *Il6* and downregulation of canonical inflammatory chemokines is visible across Tet2 KO samples.

Stemness Quiescence Pathway: Z-score Expression Tet2-KO vs WT LSK Cells

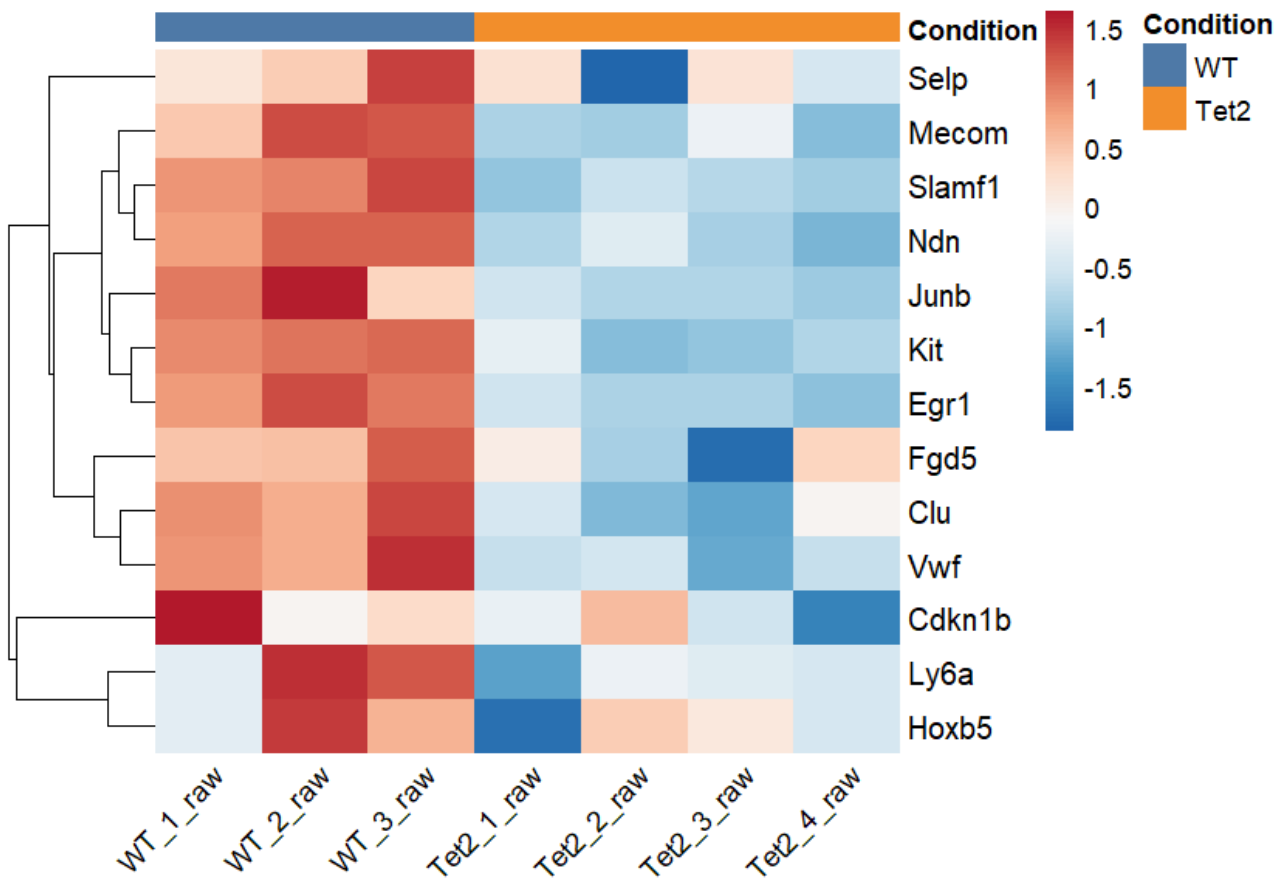


Figure 22. Heatmap of HSC stemness and quiescence genes. Downregulation of *Gata2*, *Hlf*, and *FoxO*-target genes in Tet2 KO samples confirms loss of the quiescent HSC transcriptional program.

Section 8 -- Target Gene Panel

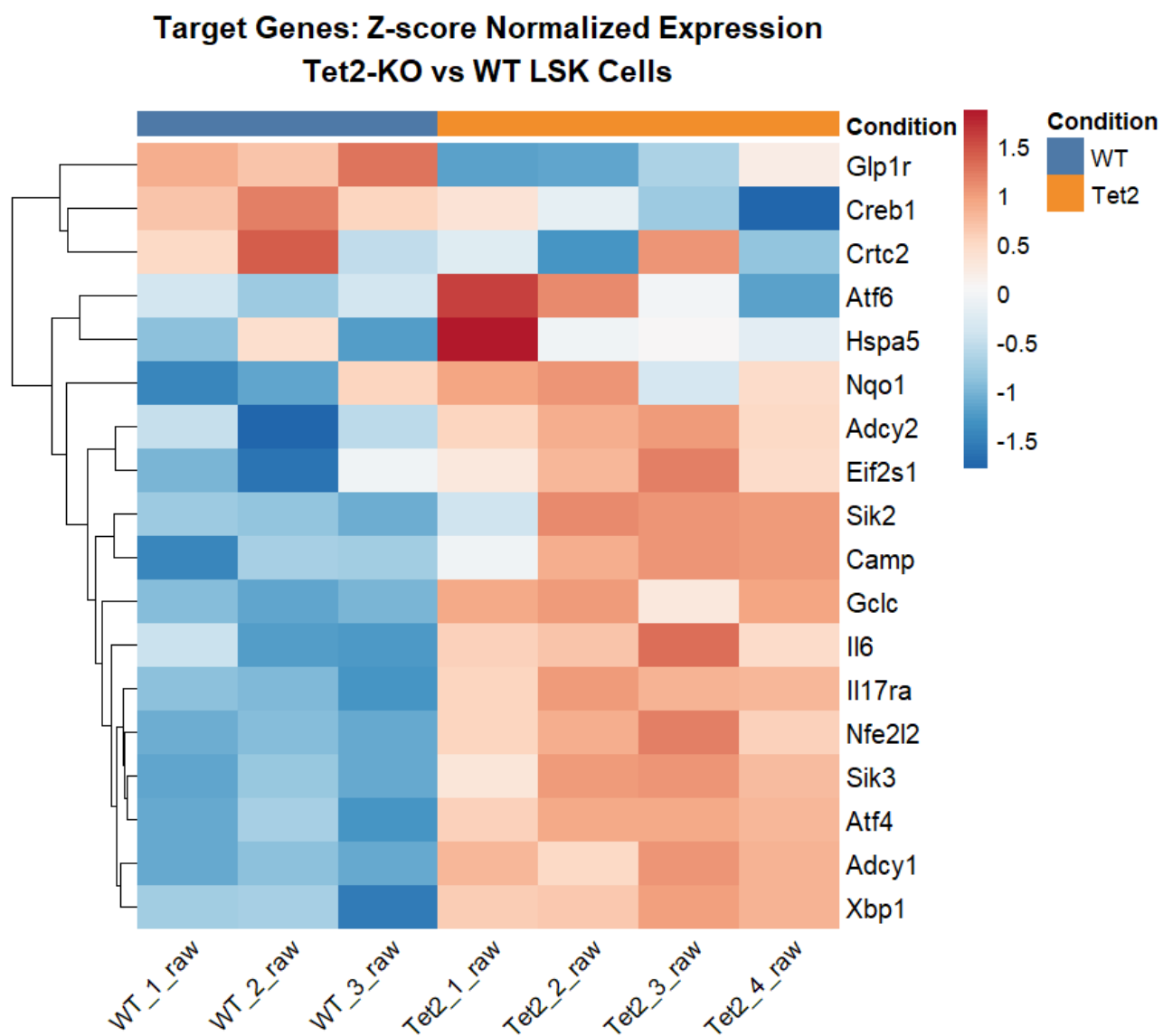


Figure 23. Heatmap of curated target genes including *Il17ra*, *Nfe2l2*, *Sik3*, *Atf4*, and *Il6*. Clear separation between WT and Tet2 KO is observed for most target genes.

Section 8 -- CT Target Gene Panel

Target Genes: Z-score Normalized Expression Tet2-KO vs WT LSK Cells

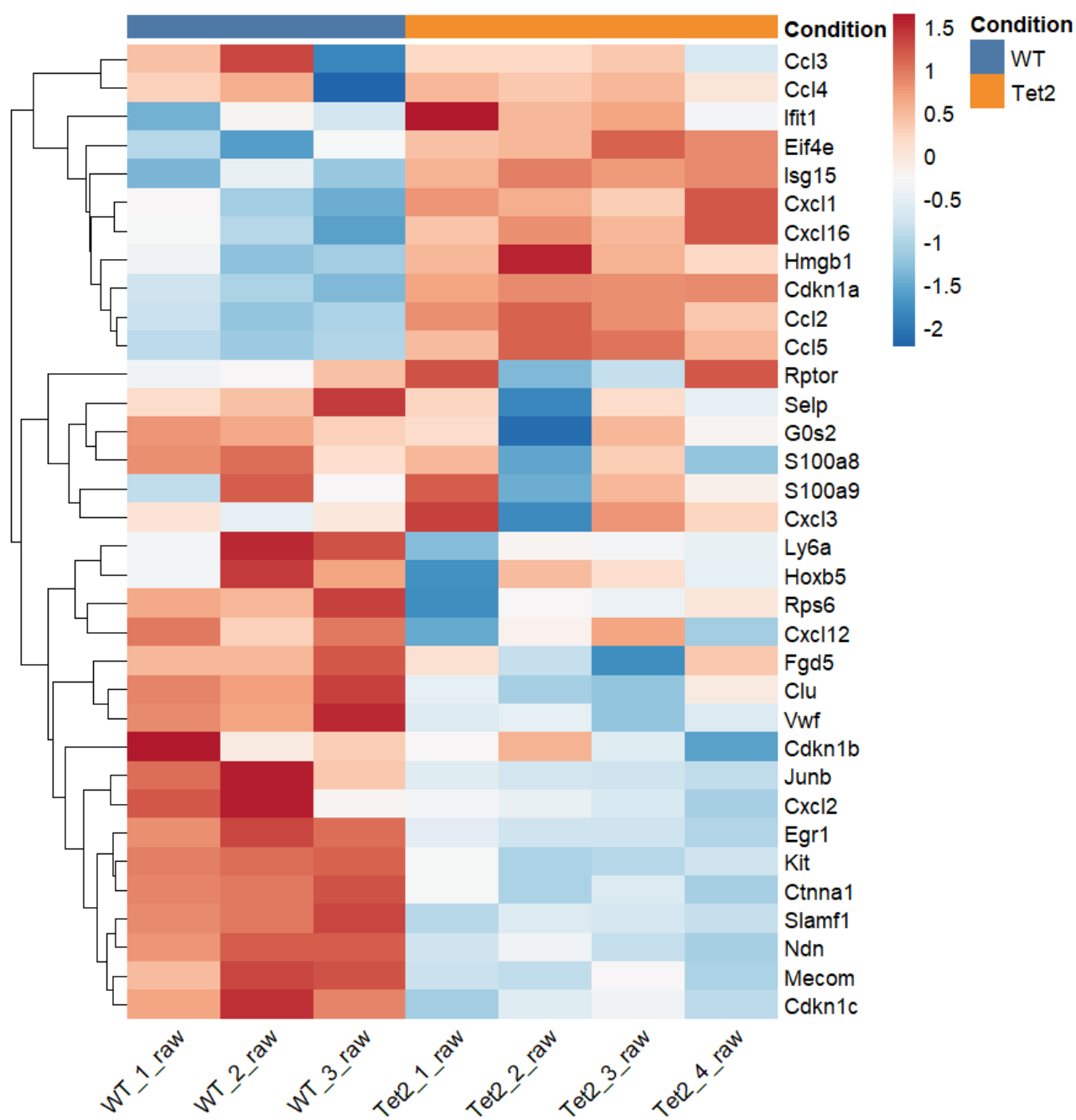


Figure 24. Heatmap of cytotoxin/CT target genes. Differential expression of this panel in Tet2 KO provides insight into altered cytotoxic and inflammatory gene regulation.

METHODS SUMMARY 9

Input Data:

Mouse WT vs Tet2 LSK (Lin- Sca1+ cKit+) bulk RNA-seq count data.

Differential Expression:

DESeq2 (R/Bioconductor). Significance threshold: $\text{padj} < 0.05$, $|\log_2\text{FC}| > 1$ for volcano classification. Wald test with Benjamini-Hochberg correction for multiple testing.

GSEA -- Hallmark:

clusterProfiler / fgsea with MSigDB Hallmark gene sets (H collection, v2023.1). Ranked by $\log_2\text{FC} \times -\log_{10}(\text{pvalue})$ signed statistic. padj threshold < 0.05 .

GSEA -- KEGG:

clusterProfiler with KEGG Mus musculus (mmu) gene sets. Gene IDs mapped to Entrez. Permutation-based NES with BH-adjusted p-values. padj threshold < 0.05 .

GSEA -- GO:

Gene Ontology enrichment (Biological Process) using clusterProfiler enrichGO. Results visualized as dotplot and enrichment map.

ORA -- Direction Split:

Genes split by direction (up/down in Tet2 KO at $\text{padj} < 0.05$, $|\log_2\text{FC}| > 1$). enrichKEGG applied separately to upregulated and downregulated gene sets. 11 selected immune/hematopoietic pathways plotted for directional comparison.

Visualization:

ggplot2, EnhancedVolcano, pheatmap, clusterProfiler plot utilities. Heatmaps generated for curated gene sets: inflammatory, cytokine, exhaustion, stemness/quiescence, and target gene panels.

Report generated programmatically from DESeq2 output, GSEA result tables, ORA results, and figure PNGs stored in DEG_Analysis_Results/. For questions contact the Santhosh Lab.